1) Test-Connection and nslookup

```
PS C:\Users\nivesh> Test-Connection www.google.com
>>

Source      Destination     IPV4Address     IPV6Address                     Bytes    Time(ms)
------      -----------     -----------     -----------                     -----    --------
DESKTOP     www.google.com  142.251.221.100                                   32       23
DESKTOP     www.google.com  142.251.221.100                                   32       43
DESKTOP     www.google.com  142.251.221.100                                   32       104
DESKTOP     www.google.com  142.251.221.100                                   32       24


PS C:\Users\nivesh>
PS C:\Users\nivesh>
PS C:\Users\nivesh>
PS C:\Users\nivesh>
PS C:\Users\nivesh> Test-Connection www.facebook.com
>>

Source      Destination     IPV4Address     IPV6Address                     Bytes    Time(ms)
------      -----------     -----------     -----------                     -----    --------
DESKTOP     www.facebook... 57.144.208.1                                     32       21
DESKTOP     www.facebook... 57.144.208.1                                     32       170
DESKTOP     www.facebook... 57.144.208.1                                     32       23
DESKTOP     www.facebook... 57.144.208.1                                     32       26


PS C:\Users\nivesh> Test-Connection www.amazon.com
>>

Source      Destination     IPV4Address     IPV6Address                     Bytes    Time(ms)
```

```
------      -----------    -----------    -----------                    -----   --------
DESKTOP     www.amazon.com  108.159.19.102                                32      101
DESKTOP     www.amazon.com  108.159.19.102                                32      25
DESKTOP     www.amazon.com  108.159.19.102                                32      21
DESKTOP     www.amazon.com  108.159.19.102                                32      33


PS C:\Users\nivesh> Test-Connection www.github.com
>>

Source      Destination    IPV4Address    IPV6Address                    Bytes   Time(ms)
------      -----------    -----------    -----------                    -----   --------
DESKTOP     www.github.com  20.207.73.82                                 32      40
DESKTOP     www.github.com  20.207.73.82                                 32      56
DESKTOP     www.github.com  20.207.73.82                                 32      93
DESKTOP     www.github.com  20.207.73.82                                 32      195


PS C:\Users\nivesh> Test-Connection www.cisco.com
>>

Source      Destination    IPV4Address    IPV6Address                    Bytes   Time(ms)
------      -----------    -----------    -----------                    -----   --------
DESKTOP     www.cisco.com   23.60.108.111                                32      37
DESKTOP     www.cisco.com   23.60.108.111                                32      72
DESKTOP     www.cisco.com   23.60.108.111                                32      85
DESKTOP     www.cisco.com   23.60.108.111                                32      38


PS C:\Users\nivesh> nslookup www.cisco.com
>>
```

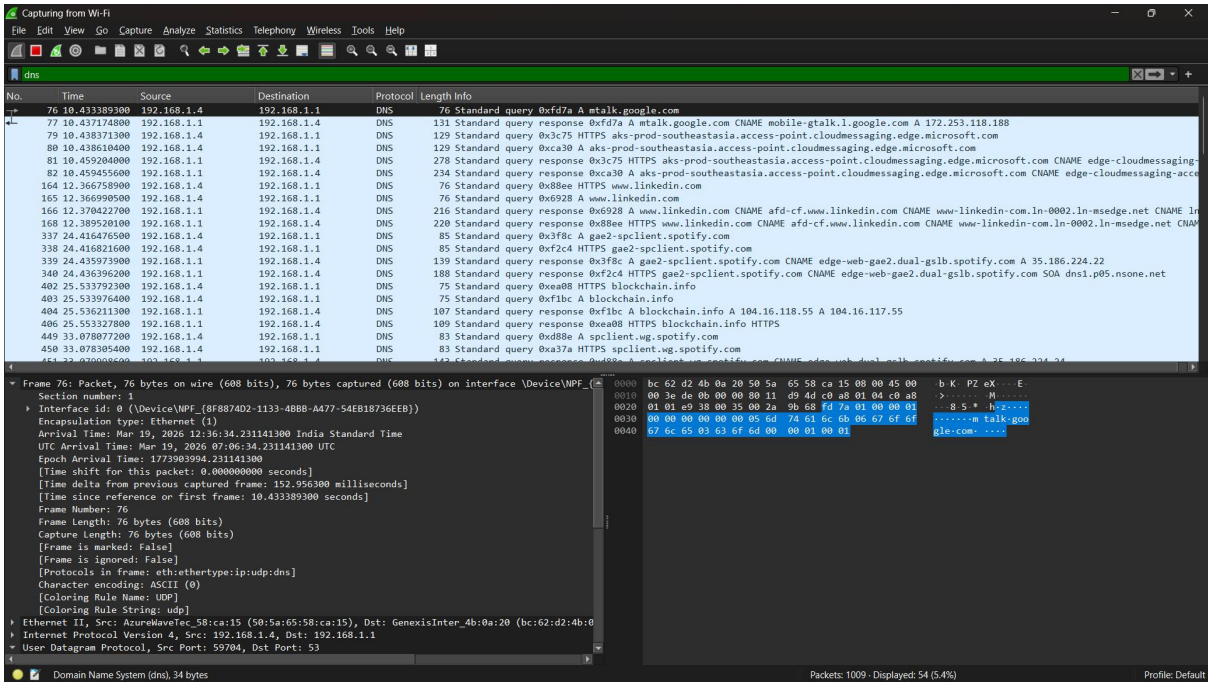
```
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    e2867.dsca.akamaiedge.net
Addresses:  2600:1417:53:193::b33
          2600:1417:53:1a9::b33
          23.60.108.111
Aliases:  www.cisco.com
          www.cisco.com.akadns.net
          wwwds.cisco.com.edgekey.net
          wwwds.cisco.com.edgekey.net.globalredir.akadns.net

PS C:\Users\nivesh> nslookup www.github.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    github.com
Address:  20.207.73.82
Aliases:  www.github.com

PS C:\Users\nivesh> nslookup www.amazon.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    e15316.dsca.akamaiedge.net
Addresses:  2600:1417:53:1ab::3bd4
```

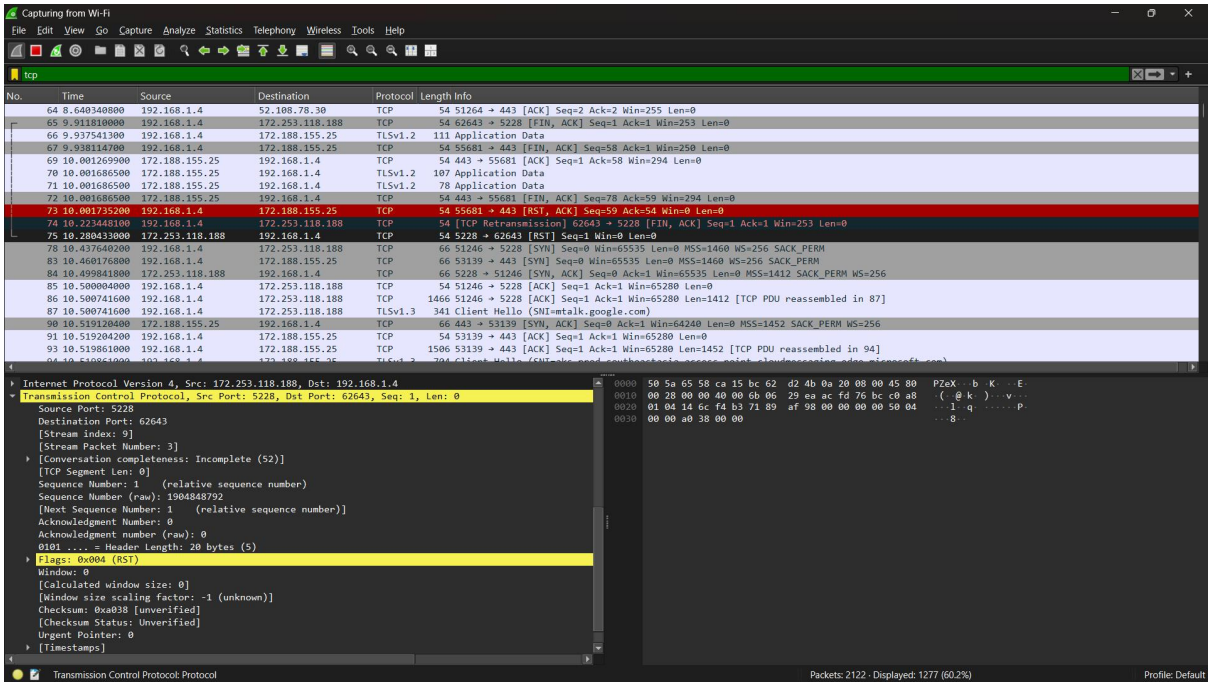
```
        2600:1417:53:1bb::3bd4
        23.60.110.84
Aliases:  www.amazon.com
        tp.47cf2c8c9-frontier.amazon.com
        www.amazon.com.edgekey.net

PS C:\Users\nivesh> nslookup www.facebook.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    star-mini.c10r.facebook.com
Addresses:  2a03:2880:f368:1:face:b00c:0:25de
        57.144.208.1
Aliases:  www.facebook.com

PS C:\Users\nivesh> nslookup www.google.com
>>
Server:  UnKnown
Address:  10.126.144.86

Non-authoritative answer:
Name:    www.google.com
Addresses:  2404:6800:4007:802::2004
        142.251.150.119
        142.251.151.119
        142.251.152.119
        142.251.153.119
        142.251.154.119
        142.251.155.119
```

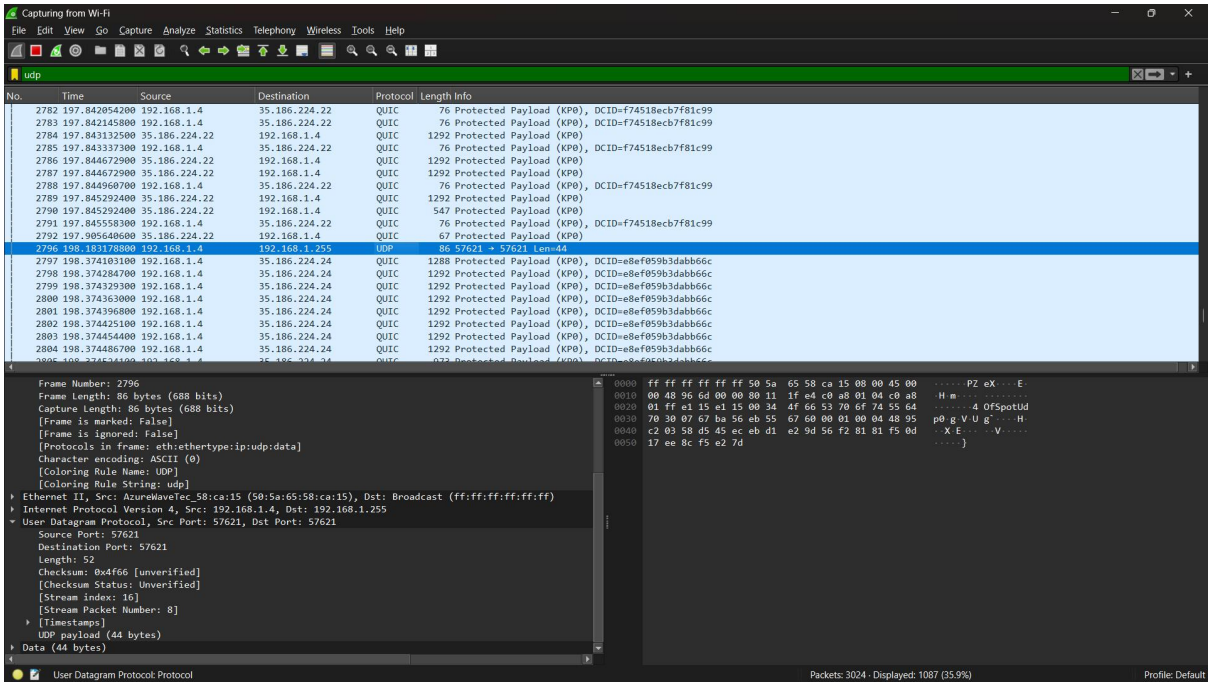
142.251.156.119

142.251.157.119

2)







3) nivesh@DESKTOP:~$ traceroute google.com

traceroute to google.com (142.251.220.110), 30 hops max, 60 byte packets

1 DESKTOP.mshome.net (172.23.96.1) 0.472 ms 0.433 ms 0.425 ms

2 192.168.1.1 (192.168.1.1) 5.797 ms 5.292 ms 5.979 ms

3 172.17.0.1 (172.17.0.1) 9.964 ms 9.947 ms 11.783 ms

4 * * *

5 172.16.178.1 (172.16.178.1) 30.287 ms 31.153 ms 31.817 ms

6 103.48.181.250 (103.48.181.250) 23.113 ms 21.978 ms 29.948 ms

7 * * *

8 * * *

9 142.250.235.104 (142.250.235.104) 24.582 ms 142.251.55.62 (142.251.55.62) 19.650 ms 142.251.55.224 (142.251.55.224) 26.060 ms

10 142.250.208.230 (142.250.208.230) 19.616 ms 142.251.55.223 (142.251.55.223) 20.151 ms 20.103 ms

11 hkg07s52-in-f14.1e100.net (142.251.220.110) 20.657 ms 172.253.70.167 (172.253.70.167) 29.432 ms pnmaaa-be-in-f14.1e100.net (142.251.220.110) 18.662 ms

nivesh@DESKTOP:~$ traceroute embedur.ai

traceroute to embedur.ai (162.159.137.54), 30 hops max, 60 byte packets

1 DESKTOP.mshome.net (172.23.96.1) 0.680 ms 0.585 ms 0.565 ms

2 192.168.1.1 (192.168.1.1) 3.105 ms 3.083 ms 2.028 ms

3 172.17.0.1 (172.17.0.1) 8.704 ms 15.109 ms 11.150 ms

4 160.30.102.74 (160.30.102.74) 38.252 ms 39.216 ms *

5 172.16.178.1 (172.16.178.1) 37.054 ms 36.175 ms 36.648 ms

6 103.48.181.250 (103.48.181.250) 35.298 ms 32.985 ms 33.170 ms

7 nsg-static-041.158.72.182.airtel.in (182.72.158.41) 35.977 ms 37.778 ms 23.895 ms

8 116.119.36.60 (116.119.36.60) 19.742 ms 25.609 ms 25.318 ms

9 * * *

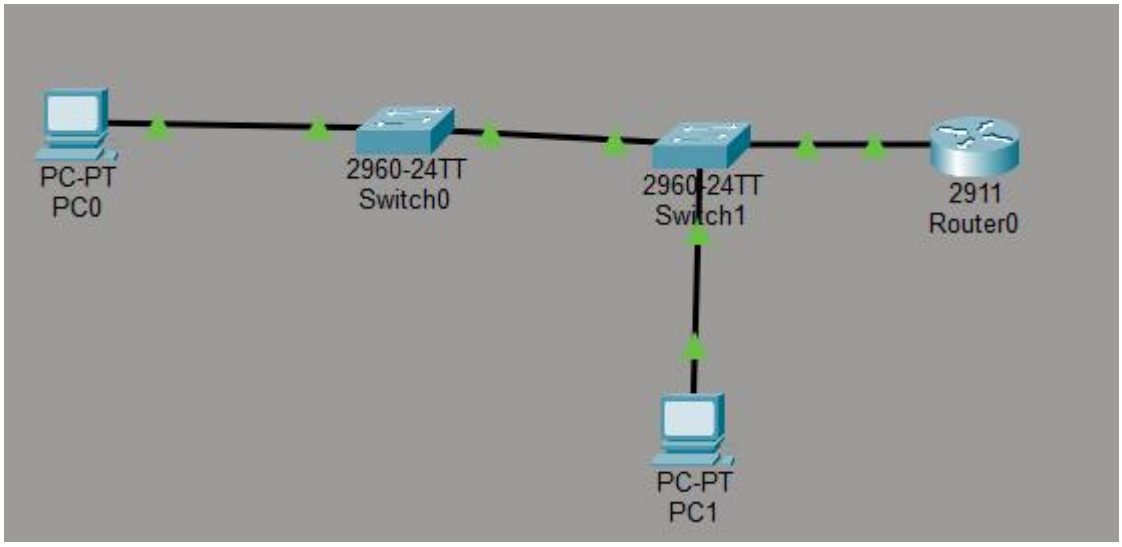
10 * 182.79.161.171 (182.79.161.171) 28.714 ms *

11 162.158.52.19 (162.158.52.19) 24.937 ms 162.158.52.35 (162.158.52.35) 72.686 ms 72.552 ms

12 162.159.137.54 (162.159.137.54) 22.948 ms 19.008 ms 19.181 ms

nivesh@DESKTOP:~$

5) Trunk Between Switches



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time=6ms TTL=128
Reply from 192.168.10.2: bytes=32 time=5ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 6ms, Average = 3ms

C:\>
```

Switch 0 code

Switch>enable

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#interface f0/1

Switch(config-if)#^Z

Switch#

%SYS-5-CONFIG_I: Configured from console by console

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#vlan 10

Switch(config-vlan)#vlan 20

Switch(config-vlan)#interface f0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 10

Switch(config-if)#interface f0/2

Switch(config-if)#switchport mode trunk

Switch 1 Code

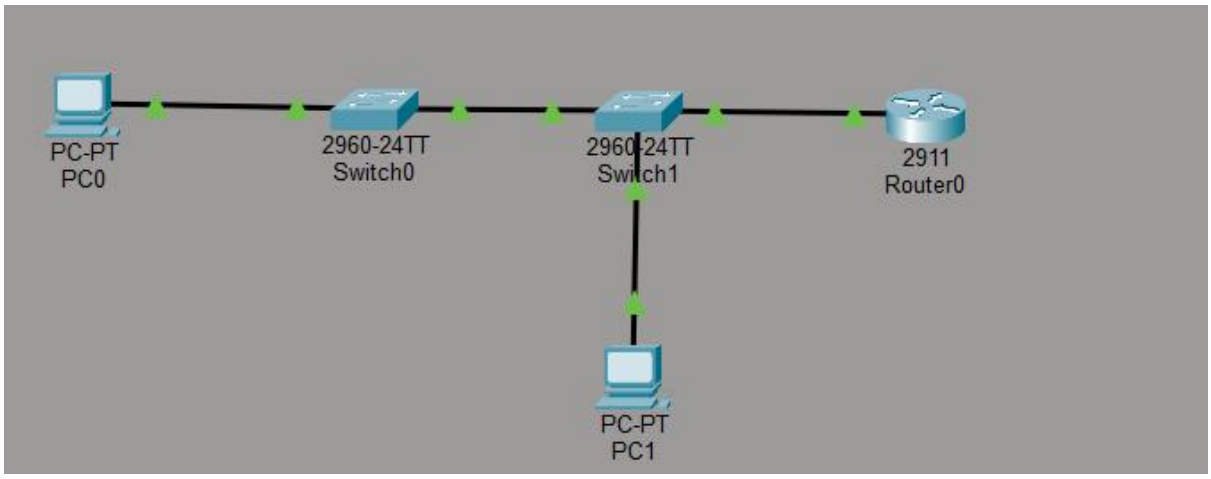
```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/1
Switch(config-if)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#vlan 20
Switch(config-vlan)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface f0/2
Switch(config-if)#switchport mode trunk
```

Router 0 Code

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#ip address 192.168.10.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g/0.10
                            ^
% Invalid input detected at '^' marker.

Router(config)#interface g0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1
% Incomplete command.
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#interface g0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#end
```

6) Change native vlan



Switch>show interfaces trunk

Port Mode Encapsulation Status Native vlan

Fa0/2 on 802.1q trunking 99

Port Vlans allowed on trunk

Fa0/2 1-1005

Port Vlans allowed and active in management domain

Fa0/2 1,10,20,99

Port Vlans in spanning tree forwarding state and not pruned

Fa0/2 1,10,20,99

VLAN Name Status Ports

---- --------- --------- ---------

1 default active Fa0/4, Fa0/5, Fa0/6, Fa0/7

Fa0/8, Fa0/9, Fa0/10, Fa0/11

Fa0/12, Fa0/13, Fa0/14, Fa0/15

Fa0/16, Fa0/17, Fa0/18, Fa0/19

Fa0/20, Fa0/21, Fa0/22, Fa0/23

Fa0/24, Gig0/1, Gig0/2

10 VLAN0010 active

20 VLAN0020 active Fa0/3

99 VLAN0099 active

1002 fddi-default active

1003 token-ring-default active

1004 fddinet-default active

1005 trnet-default active

Switch#show interfaces trunk

Port Mode Encapsulation Status Native vlan

Fa0/1 on 802.1q trunking 99

Fa0/2 on 802.1q trunking 99

Port Vlans allowed on trunk

Fa0/1 1-1005

Fa0/2 1-1005

Port Vlans allowed and active in management domain

Fa0/1 1,10,20,99

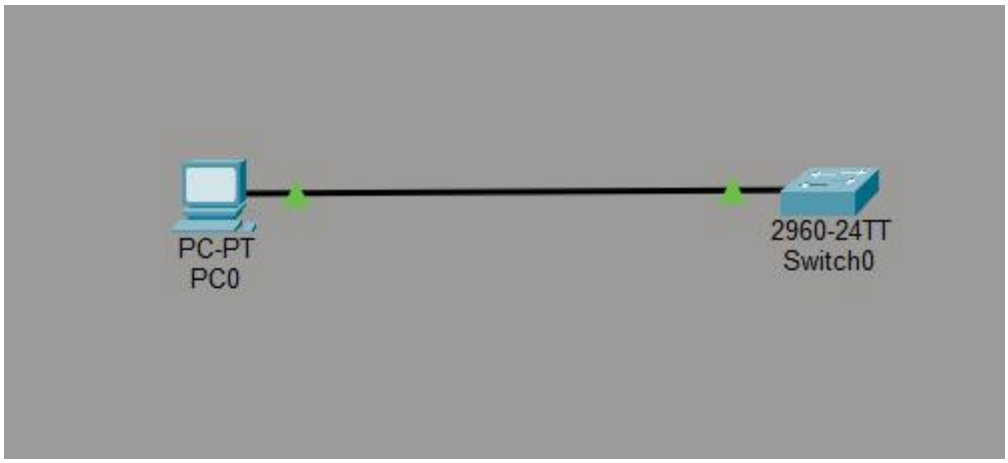
Fa0/2 1,10,20,99

Port Vlans in spanning tree forwarding state and not pruned

Fa0/1 1,10,20,99

Fa0/2 1,10,20,99

7) Telnet to Switch



SWITCH CODE INCLUDING TELNET SETUP

Switch>enable

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#vlan 99

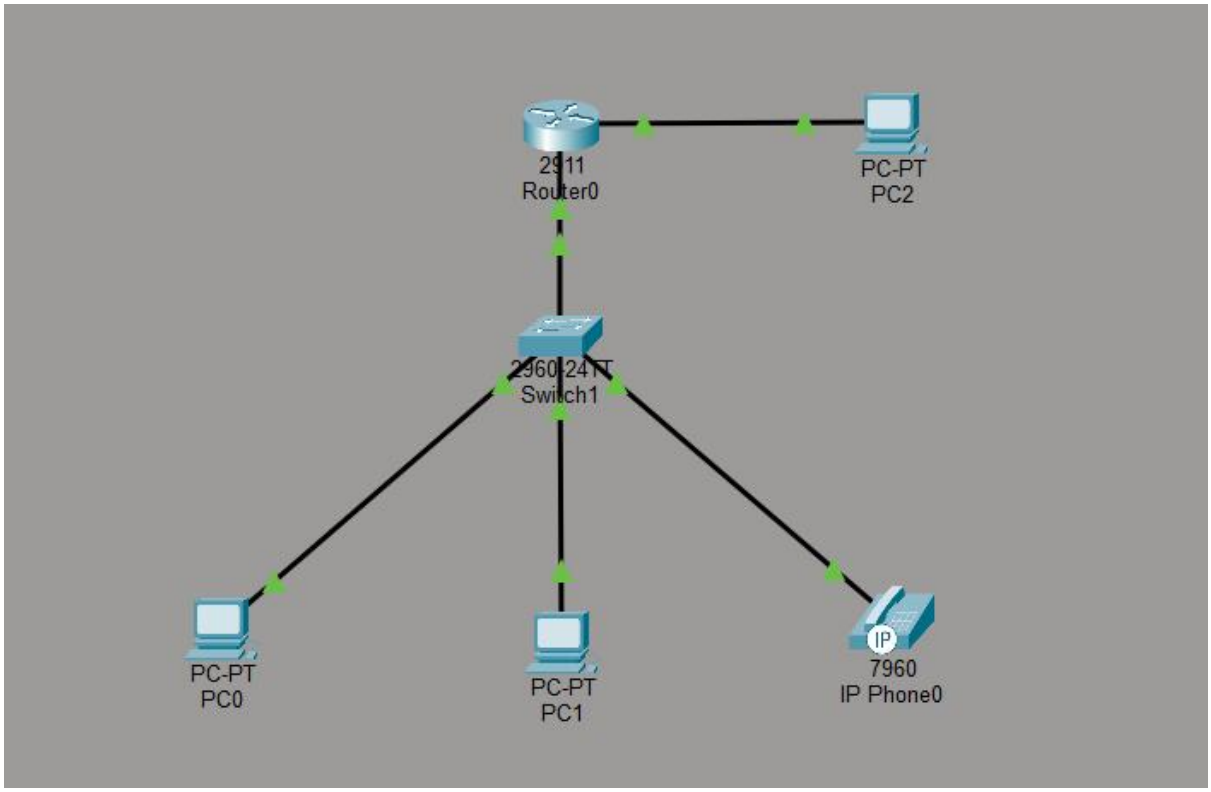
```
Switch(config-vlan)#name MANAGEMENT
Switch(config-vlan)#exit
Switch(config)#interface vlan 99
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

Switch(config-if)#ip address 192.168.99.2 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#interface f0/1
^
% Invalid input detected at '^' marker.
Switch#interface fa0/1
^
% Invalid input detected at '^' marker.
Switch#enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 99
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
no
% Incomplete command.
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#end
```

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console


Switch#ip default-gateway 192.168.99.1
^
% Invalid input detected at '^' marker.
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#ip default-gateway 192.168.99.1
Switch(config)#line vty 0 4
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#transport input telnet
Switch(config-line)#hostname Switch
Switch(config)#ip domain-name lab.local
Switch(config)#username admin secret cisco123
Switch(config)#crypto key generate rsa
% Please define a hostname other than Switch.
Switch(config)#crypto key generate rsa 1024
^
% Invalid input detected at '^' marker.
Switch(config)#crypto key generate rsa
% Please define a hostname other than Switch.
Switch(config)#crypto key generate rsa 1024
^
% Invalid input detected at '^' marker.
Switch(config)#conf t
^
% Invalid input detected at '^' marker.
Switch(config)#hostname SW1
SW1(config)#ip domain-name lab.local
```

```
SW1(config)#crypto key generate rsa
The name for the keys will be: SW1.lab.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

SW1(config)#line vty 0 4
*Mar 1 0:7:33.810: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW1(config-line)#login local
SW1(config-line)#transport input ssh telnet
^
% Invalid input detected at '^' marker.
SW1(config-line)#exit
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#write memory
Building configuration...
[OK]
SW1#
SW1#
```

```
C:\>ping 192.168.99.2

Pinging 192.168.99.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.99.2: bytes=32 time<1ms TTL=255
Reply from 192.168.99.2: bytes=32 time<1ms TTL=255
Reply from 192.168.99.2: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.99.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet 192.168.99.2
Trying 192.168.99.2 ...Open


User Access Verification

Username:
% Username:  timeout expired!

[Connection to 192.168.99.2 closed by foreign host]
C:\>telnet 192.168.99.2
Trying 192.168.99.2 ...Open


User Access Verification

Username: admin
Password:
SW1>ssh -l admin 192.168.99.2

Password:

Password:

Password:

[Connection to 192.168.99.2 closed by foreign host]
SW1>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual down                  down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
```

8)Setting up different vlans for pc and VOIP mobile



```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name DATA
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name VOICE
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name MGMT
Switch(config-vlan)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface fa0/3
Switch(config-if)#switchport access vlan 10
Switch(config-if)#switchport mode access
Switch(config-if)#switchport voice vlan 20
Switch(config-if)#interface fa0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#interface vlan 99
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan99, changed state to up

Switch(config-if)#ip address 192.168.99.2 255.255.255.
                                          ^
% Invalid input detected at '^' marker.

Switch(config-if)#ip address 192.168.99.2 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#ip default-gateway 192.168.99.1
Switch(config)#
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan99, changed state to up
```

```
Router>
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#interface g0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface g0/0.99
Router(config-subif)#encapsulation dot1Q 99 native
Router(config-subif)#ip address 192.168.99.1 255.255.255.0
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.99, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.99, changed state to up

Router(config-subif)#interface g0/1
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#interface g0/0
Router(config-if)#ip nat inside
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip nat outside
Router(config-if)#access-list 1 permit 192.168.0.0 0.0.255.255
Router(config)#ip nat inside source list 1 interface g0/1 overload
Router(config)#
```

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.1.1.2

Pinging 200.1.1.2 with 32 bytes of data:

Request timed out.
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127
Reply from 200.1.1.2: bytes=32 time=1ms TTL=127
Reply from 200.1.1.2: bytes=32 time<1ms TTL=127

Ping statistics for 200.1.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
Switch>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   DATA                             active    Fa0/1, Fa0/2, Fa0/3
20   VOICE                            active    Fa0/3
99   MGMT                             active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch>
```

```
Switch>show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/4       on           802.1q         trunking      99

Port        Vlans allowed on trunk
Fa0/4       1-1005

Port        Vlans allowed and active in management domain
Fa0/4       1,10,20,99

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/4       1,10,20,99

Switch>
```

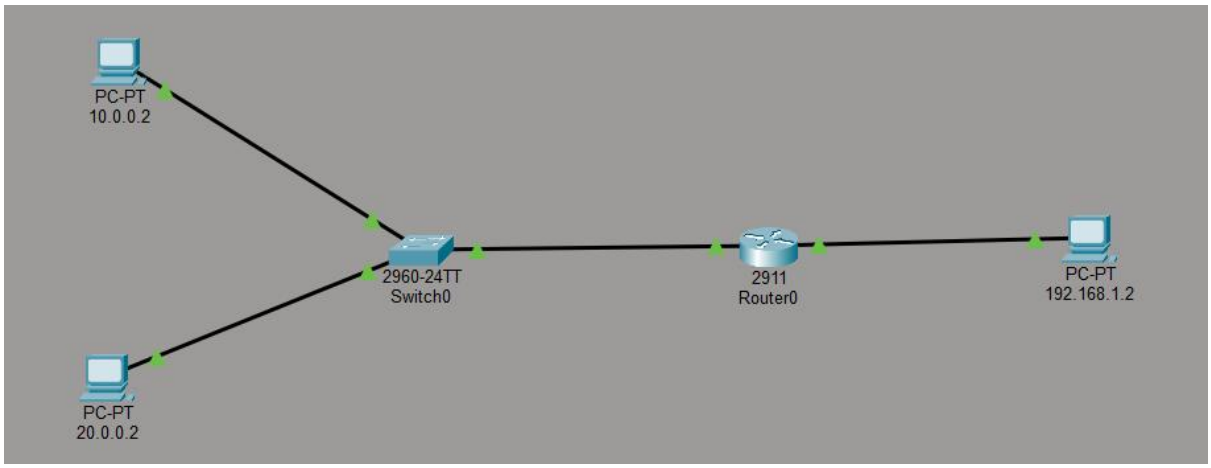
8)

The lack of communication between VLAN 10 and VLAN 20 is expected behavior because VLANs operate as separate Layer 2 broadcast domains, meaning devices in different VLANs cannot exchange traffic directly through a switch. Even if both PCs are correctly assigned to their respective VLANs, the switch will only forward frames within the same VLAN and will not perform any Layer 3 routing between them. To enable communication, an external routing mechanism such as a router configured with inter-VLAN routing (commonly using router-on-a-stick) or a Layer 3 switch with switched virtual interfaces (SVIs) must be implemented so that traffic can be routed between VLAN 10 and VLAN 20.

9)

In a network where VLAN 10 and VLAN 20 are configured on a switch, devices in different VLANs cannot communicate with each other by default because VLANs operate at Layer 2 (Data Link Layer) and isolate broadcast domains. This means that traffic from one VLAN is not forwarded to another VLAN unless a Layer 3 device is introduced. The issue arises when there is no inter-VLAN routing configured, even though VLANs and ports are correctly assigned. To resolve this problem, a router or Layer 3 switch must be used to perform Inter-VLAN Routing, typically through methods like Router-on-a-Stick, where subinterfaces are created for each VLAN and traffic is routed between them. Once routing is enabled and proper gateway addresses are configured on the end devices, communication between VLAN 10 and VLAN 20 becomes possible.

10)INTER VLAN ROUTING



SWITCH CODE

enable

conf t

vlan 10

name VLAN10

vlan 20

name VLAN20

```
exit
interface fa0/1
switchport mode access
switchport access vlan 10
no shutdown
interface fa0/2
switchport mode access
switchport access vlan 20
no shutdown
interface fa0/3
switchport mode trunk
no shutdown
end
write memory
```

Router Code

```
enable
conf t
interface g0/0
no shutdown
interface g0/0.10
encapsulation dot1Q 10
ip address 10.0.0.1 255.255.255.0
ip nat inside
interface g0/0.20
encapsulation dot1Q 20
ip address 20.0.0.1 255.255.255.0
ip nat inside
interface g0/1
ip address 192.168.1.1 255.255.255.0
ip nat outside
```

no shutdown

access-list 1 permit 10.0.0.0 0.0.0.255

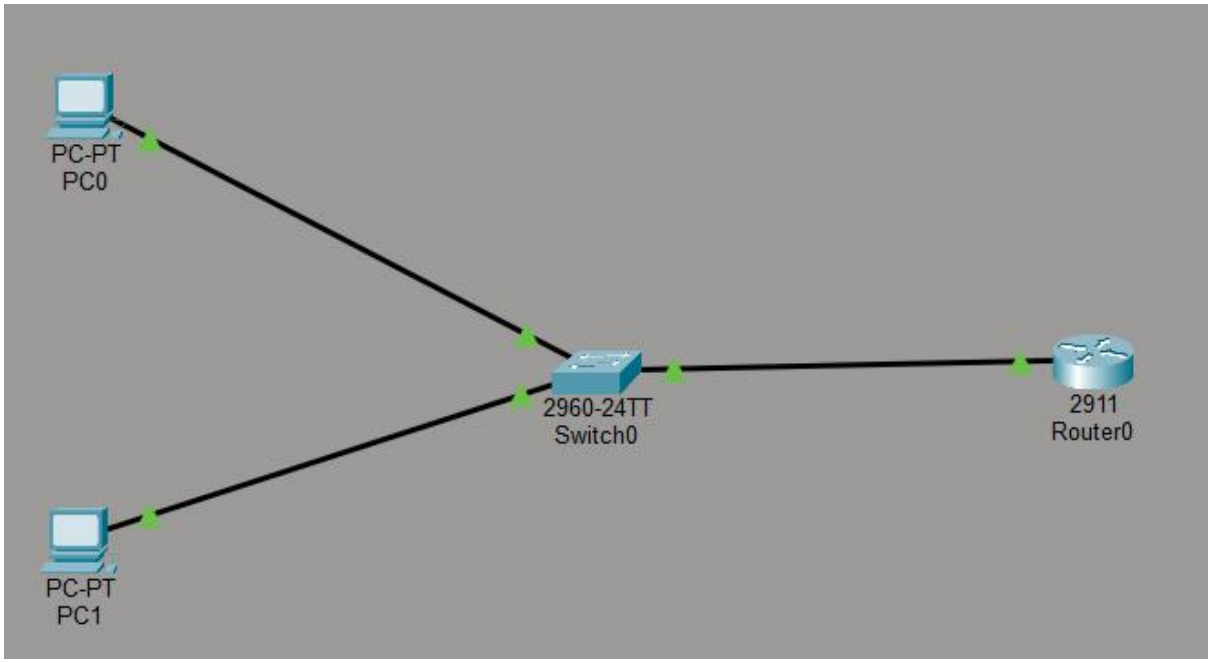
access-list 1 permit 20.0.0.0 0.0.0.255

ip nat inside source list 1 interface g0/1 overload

end

write memory

11) Blocking Traffic Between 2 PCS



Router Code

Router>

Router>enable

Router#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface g0/0

Router(config-if)#no shutdown

Router(config-if)#

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

```
Router(config-if)#interface g0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.1.1 255
^
% Invalid input detected at '^' marker.
Router(config-subif)#ip address 192.168.1.1 255.255
^
% Invalid input detected at '^' marker.
Router(config-subif)#ip address 192.168.1.1 255.255.255.0
Router(config-subif)#interface g0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#
Router(config-subif)#exit
Router(config)#ip access-list extend GC
Router(config-ext-nacl)#deny ip host 192.168.1.10 host 192.168.2.10
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#interface g0/0.10
Router(config-subif)#ip access-group GC in
```

Router(config-subif)#show access-lists

^

% Invalid input detected at '^' marker.

Router(config-subif)#exit

Router(config)#exit

Router#

%SYS-5-CONFIG_I: Configured from console by console

Router#show access-lists

Extended IP access list GC

10 deny ip host 192.168.1.10 host 192.168.2.10 (4 match(es))

20 permit ip any any

Switch Code

```
Switch>
Switch>enable
Switch#con f
% Ambiguous command: "con f"
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#exit
Switch(config)#interface f0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#interface f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#interface f0/3
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
```

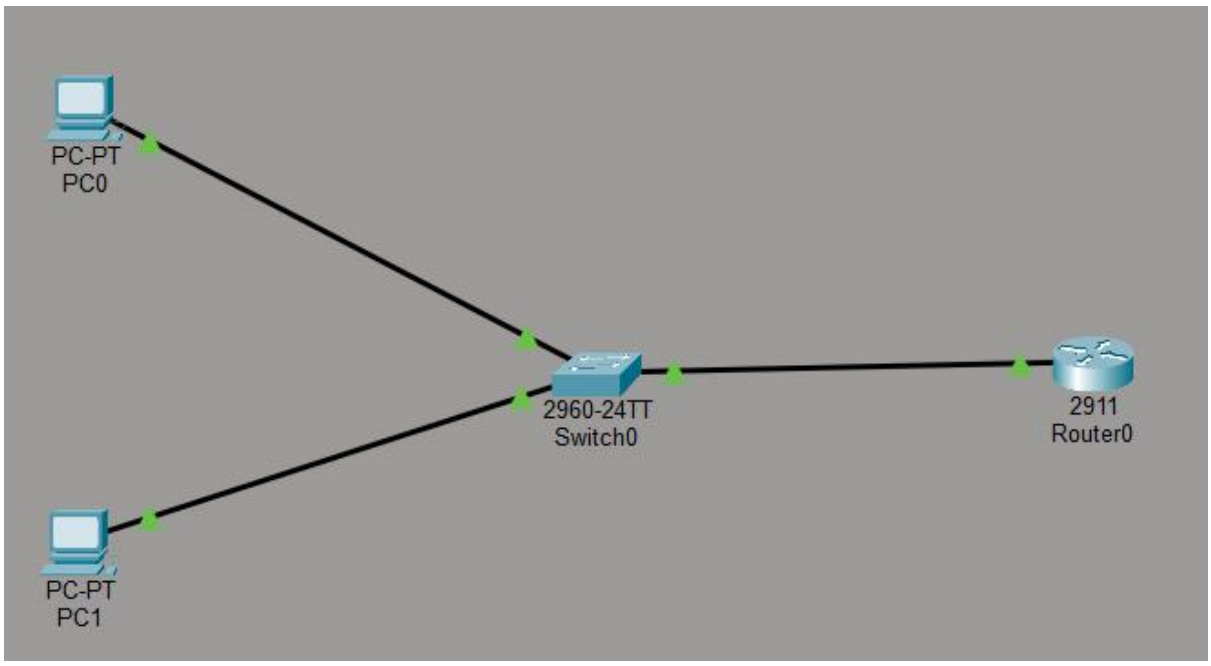
BEFORE CONFIGURING ACL

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

AFTER CONFIGURING ACL

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
C:\>
```

12)



Router CODE added

Router(config)#no ip access-list extended GC

Router(config)#ip access-list extended GV

Router(config-ext-nacl)#deny ip 192.168.1.0 0.0.0.255 host 192.168.2.10

Router(config-ext-nacl)#permit ip any any

Router(config-ext-nacl)#interface g0/0.10

Router(config-subif)#ip access-list GV in

^

% Invalid input detected at '^' marker.

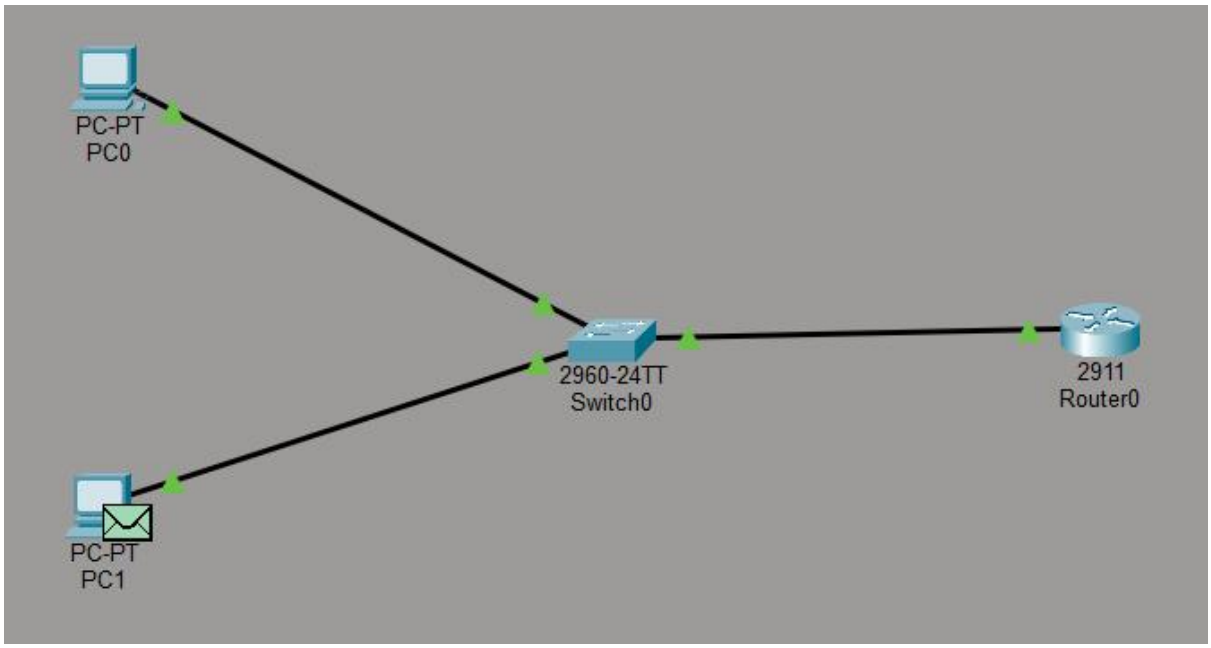
Router(config-subif)#ip access-group GV in

Router(config-subif)#

Router#

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

C:\>
```

13)



Router(config)#access-list 100 deny tcp host 192.168.1.10 host 192.168.2.10 eq 80

Router(config)#access-list 100 permit ip any any

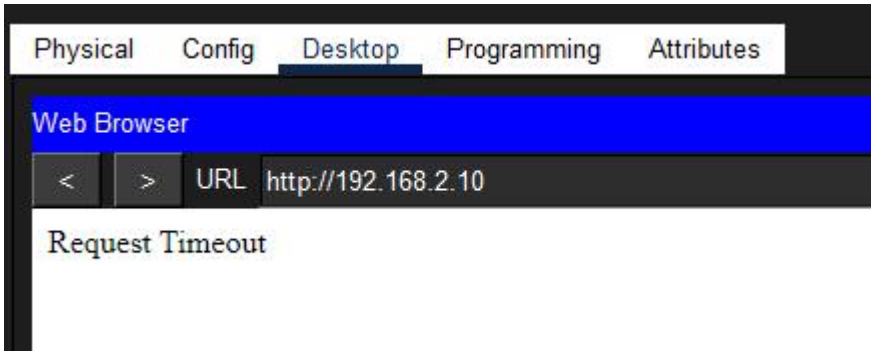
Router(config)#interface g0/0.10

Router(config-subif)#ip access-group 100 in

Router(config-subif)#no ip access-list 100

Router(config-subif)#no access-list 100

Router(config)#



15) **IPERF FROM WSL TO CMD (LOCALHOST)**

WSL OP

vignesh@DESKTOPGV:~$ sudo apt install iperf3 -y

vignesh@DESKTOPGV:~$ iperf3 -s

Server listening on 5201 (test #1)

Accepted connection from 172.23.96.1, port 58070

[5] local 172.23.98.200 port 5201 connected to 172.23.96.1 port 58071

[ID] Interval Transfer Bitrate

[5] 0.00-1.00 sec 743 MBytes 6.23 Gbits/sec

[5] 1.00-2.00 sec 773 MBytes 6.48 Gbits/sec

[5] 2.00-3.00 sec 772 MBytes 6.48 Gbits/sec

[5] 3.00-4.00 sec 747 MBytes 6.26 Gbits/sec

[5] 4.00-5.00 sec 698 MBytes 5.86 Gbits/sec

[5] 5.00-6.00 sec 754 MBytes 6.32 Gbits/sec

[5] 6.00-7.00 sec 744 MBytes 6.24 Gbits/sec

```
[  5]   7.00-8.00   sec   735 MBytes  6.16 Gbits/sec
[  5]   8.00-9.00   sec   748 MBytes  6.27 Gbits/sec
[  5]   9.00-9.17   sec   127 MBytes  6.31 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  5]   0.00-9.17   sec  6.68 GBytes  6.26 Gbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201 (test #2)
```

CMD OP

```
PS C:\Users\guruv\Downloads\iperf3.1.4_64> .\iperf3.exe -c 172.23.98.200
Connecting to host 172.23.98.200, port 5201
[  4] local 172.23.96.1 port 58071 connected to 172.23.98.200 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec   687 MBytes  5.76 Gbits/sec
[  4]   1.00-2.00   sec   706 MBytes  5.92 Gbits/sec
[  4]   2.00-3.00   sec   702 MBytes  5.89 Gbits/sec
[  4]   3.00-4.00   sec   709 MBytes  5.95 Gbits/sec
[  4]   4.00-5.00   sec   655 MBytes  5.49 Gbits/sec
[  4]   5.00-6.00   sec   652 MBytes  5.47 Gbits/sec
[  4]   6.00-7.00   sec   697 MBytes  5.85 Gbits/sec
[  4]   7.00-8.00   sec   668 MBytes  5.60 Gbits/sec
[  4]   8.00-9.00   sec   672 MBytes  5.64 Gbits/sec
[  4]   9.00-10.00  sec   693 MBytes  5.81 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  6.68 GBytes  5.74 Gbits/sec                  sender
[  4]   0.00-10.00  sec  6.68 GBytes  5.74 Gbits/sec                  receiver
```

iperf Done.